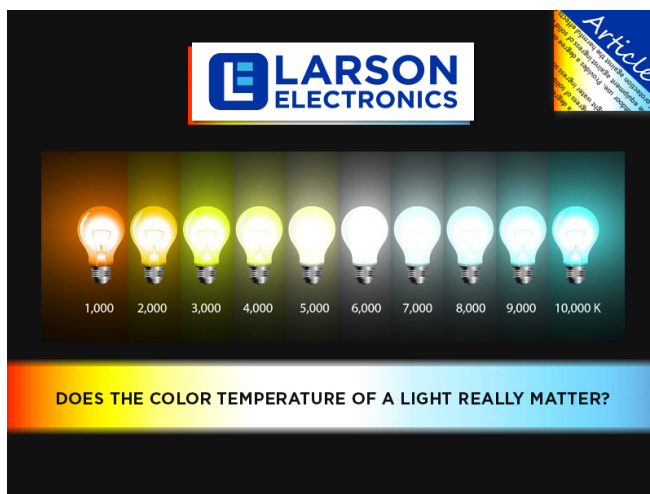


Larson Electronics

LED Lighting

Lighting 101: Color Temperature – What is the Kelvin Scale?

By Larson Electronics on December 12, 2017



What is color temperature?

Color temperature, is defined as “the temperature of an ideal black-body radiator that radiates light of a color comparable to that of the light source.” (Wikipedia) A *blackbody* is an object that absorbs all radiation, which includes visible light, infrared light, ultraviolet light, etc.



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In laymens terms, color temperature is used as a method of describing the warmth or coolness color characteristics of a light source.

Color temperature, in practice, is only useful for light sources that fall in a color spectrum of reddish/orange (yellow based) to white (blue based). This is because this spectrum corresponds closely to the radiation of a black body, whereas purples and greens do not.

This range of color can be seen when heating metal, which initially emits red light that graduates from orange to yellow, then from white to blue.

The spectrum of color temperatures is assigned numerical values, measured in degrees of Kelvin. And these values are used to describe the color emitted from fixtures when working with lighting.

Kelvin Color Temperature Scale for Light Fixtures

2000K-3000K: Gives off a warm white/yellow glow (candlelight hues)

3100K-4500K: Gives off a bright, cool white glow

4600K-6500K: Gives off a crisp daylight glow (white/bluish hues)

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Color Temperature Applications:

Color temperature has important implications for lighting. Ambient lighting will require a much different color temperature than a space dedicated to detailed tasks. For this reason, we've compiled a general guide to explain which type of space requires which color temperature, to better help you determine the correct color temperature for your application.

2700K – Warm White

This color temperature is most commonly used in homes, restaurants and hotels because of its warm, cozy and inviting glow – similar to a sunset or candlelight.

3000K – Soft White

A soft white temperature still provides warmth, but a bit more clarity for completing tasks. This color is common in bathrooms and kitchen areas.

3500K – Neutral White

A neutral white mimics natural “middle of the day” light, and is ideal for spaces that require alertness such as office spaces and retail stores. This light is still warm and easy on the eyes, but promotes focused activities.

4100K – Cool White

When an environment requires more precision, a cool white light is ideal. This lighting color is used mainly in professional garages and grocery stores, which need crisp lighting for workers and customers to see detail when working on a car, and the colors of food as correctly as possible.

5000K – Bright White

Some locations require very bright white lighting for ultimate clarity. These spaces include warehouses, sport stadiums, hospitals, ER rooms and other industries where the brightest light is needed to perform tasks correctly.

6500K – Daylight

This color temperature has an apparent blue tone, and mimics natural daylight from the sun. It is commonly used for indoor farming, greenhouses and other agricultural purposes.

As you can see the color temperature scale is extremely important when choosing a lighting fixture. Every location and environment requires different lighting tones to best suit the activity in those spaces, and the wrong color could greatly affect productivity of workers or the quality of work. To determine the color temperature you

need, consider the type of tasks that will be completed in that space, or the type of feeling you want to convey.

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